

ActInf Livestream #005.0: Context for "Multiscale integration: beyond inter

Livestream | 48:55

hello and welcome

to the active inference live stream my

name is daniel friedman and this is

going to be a

solo discussion if you're a new listener

or not

welcome to team com we are an experiment

in online team communication and

learning

related to active inference you can find

us on twitter

at inferenceactive at activeinference

gmail.com

on key base via our public team or our

youtube channel

this is a recorded and archived live

stream

so please provide us with feedback so

that we can improve our work

all backgrounds and perspectives are

welcome here

and also as far as video etiquette goes

remember to mute if there's noise in the

background

raise your hand if you have a question

and so on

here we are in act imp stream number 5.0

we're kind of exploring with the

numbering system so give us any

feedback as far as what kinds of numbers

or what kinds of shows might be helpful

for

your learning but the goal for this

discussion is to set the context for

active in

stream 5.1 in 5.1

we're going to discuss the paper

multi-scale integration

beyond internalism and externalism by

ramstead at all 2019.

the goal of this video is really to

contextualize some of the ideas

the historical threads and the

vocabulary that will help you either

read or work with the ramstead paper uh

in terms of other

fields that you might be more familiar

with or just to sort of connect ideas

and thus ground your understanding and

your practice

so i come from a different background

from the author so i have a slightly

different perspective on

perhaps even the implications of the

paper so the goal here is really just to

explore that

so that whether you listen to this by

itself

and you're looking to sort of see where

active inference is trying to relate

itself to other philosophical ideas

or whether you're coming from the active

inference side and you want to

understand some of the broader

philosophical questions

that are being addressed potentially by

active inference in both those cases i

hope this is an interesting

video for you so here are a few of the

topics that we'll be

exploring here the first one is top-down

and bottom-up causation and that's going

to be an introduction to multi-level

systems as well as thinking about
causal modeling also
next we'll talk about and raise some
questions
related to partitioning between internal
and external cognition
which is certainly the wedge or the main
issue
that the ramps that at all paper is
going to be focused on
and then we'll talk about a few
different perspectives on systems
modeling
and action that's sort of action and
inference
if you will but there's also been a lot
of philosophical thought
on systems modeling as well as action so
we can kind of see how those two fields
might come together
in the field called active inference and
next time in number 5.1
we will all come together everybody who
wants to can always get in touch with us
we will all come together
and discuss the paper to whatever extent
we've read it
whatever we thought about it any
questions that we wanted to address
any of our background knowledge uh any
of our perspectives that we bring to the
paper those are all cool things
to come about in 5.1 and if you're
listening to this after 5.1 has happened
you can either listen to 5.1 or you
could get in touch with us to be on a
future
discussion so next time in

5.1 we're going to be talking about the figures to abstract etc in this paper by ramstead at all called multi-scale integration beyond internalism and externalism and that's in the journal syntheses in 2019 so the goal of that paper and the only quotes from the paper that are going to be in this video are just from the introduction and abstract just so that we can ground the ideas more specifically in what ramps that at all frame their work as so the goal of the paper in the author's words was we focus on making explicit a description of the boundaries of cognitive systems that we think follows from taking seriously the inactive embodied and extended nature of cognition this is the idea that the boundaries of cognitive systems are nested and multiple and that with respect to its study cognition has no fixed or essential boundaries cool i think that that is really exciting and also it doesn't necessarily bring the free energy principle or active inference on too heavily it just helps understand what is the broader goal of the paper well the question that really excited me today was this

taking it seriously aspect on
multi-scale integration
it's the first two words of the paper's
title
so maybe they're trying to signal with
that the story that we want to be
working towards is multi-scale
integration so
linking things that are different
spatial and temporal scales
at once multi-scale how are we going to
do
that multi-scale integration well what
they're
provocatively stating in the title is
that we must move beyond
this debate of internalism versus
externalism which i'll contextualize
soon
and the way that they're going to take
it seriously is using a mathematical
framework
called free energy principle and it's
related process theory
active inference so that's what taking
it seriously looks like for them
especially drawing on their background
as cognitive science
experts they're really framing this as
about multi-scale cognition
and i came from a not a cognitive
sciences background
more of an evolutionary biology
background and so
multi-scale integration in the ant
colony or in the insect brain or in the
cell
there's a different set of philosophical

baggage that comes along with it
sometimes if i could characterize it so
clearly would be the focus on humans
tends to focus on the experience
and the internal view since we are human
so at the very least it's one of the
ways that we can study
emergencies by ourself as an emergent
process whereas in
computers ant colonies the cell there's
often
a tendency following let's just say
dennit
and others to imagine it as an automata
or just sort of a robot
and therefore devoid of experience or
specularity
sorry speculatively having experience
but sort of on the sci-fi fringe and
anyways when you deal with the system
you have to be in that intentional
stance
so those types of systems tend to have
perspectives that are devoid of agency
whereas human systems the debate is just
about that
is about our own agency but we're all
part of the same multi-skill integrated
framework and so we want to have a story
that works for cells that are in our gut
that are in the uh in our gut lumen in
the microbiome why should it matter why
should we have a different theory of
biology for one side of that over the
other especially when we know that
there's
important transfer between them for
example so

the authors took it seriously and they wrote this 2019 paper as well as other papers

i'm gonna try to take this seriously from

the work in biology that i have learned about

over the years and that's what taking it seriously and contextualizing looks like to me so i just thought i could start with that

here's a paper from 2012 by dennis noble who's a great systems biologist called a theory of biological relativity no privileged level of causation so you can kind of see

where he's going which is taking the idea of relativity

whether that's gravitational relativity something related to quantum mechanics for example

or just uh inter-subjective relativity like focusing on the relations that define something or the interactions that define a person

rather than an essentialist take so noble is sort of taking a broad understanding of relativity and saying well if we really look at multi-scale systems and spatial and temporal cause

then there's going to be this relativity principle maybe that could apply to biology

and what i like about his figure one and i've added two wrongs at the top is that it really clearly phrases what he's arguing against

and also makes it clear um how some of these

logical errors become deployed in real systems in real analyses

so figure one is a

model of causation for what's called upward causation

and the way to read this figure is it starts at the bottom

with the smallest thing which are genes which here

is just i believe being meant as a string of dna but that's a whole other discussion

let's just say that genes uh are able to cause proteins in rna which are another scale of organization bigger moving up towards protein and rna networks up through cells tissues organs organisms

social interactions ecological technological

governance informational systems and so on so this is a

sort of multi-scale systems perspective and the

version of events that we often get is implicitly one of what noble is characterizing accurately as upward causation

one could change the spatial metaphor it could be moving downwards it could be moving inwards or outwards

but suffice to say that this is what we mean when we say upward causation is really

from small things to big things because the top down things are often seen as

big
and the things that are just on the
fabric level of reality are often seen
as small
rightly or wrongly um and
the the story when you have this type of
a
upward causation model again that
smaller things cause bigger things
is the the genes are upstream of the
behavior or
um the uh the cell damage is upstream of
the
organs so it makes sense and there's
also a related idea called supervenience
that is that
is there a relationship between changing
the smaller thing and the bigger thing
changing
so these are the kinds of questions that
come up and this is one
direction of causation however what
noble
ends up framing in this uh no
privileged level of causation model is
that yes there is
influence or flow from so-called the
upward direction so that's the genes
and uh going all the way up in
influencing things that are at larger
scale
but also in a very real and causal sense
there are top-down causation
that can be equally important for
regulating the system so even if in the
short time scale
the in uh influence of the bottom up is
extremely strong if there's no top-down

force sometimes
any amount of bottom-up power can just
cause the system to dissipate and
therefore not exist
so in that sense all systems that we
observe continuing to dissipate energy
for example like active inference
foragers informational foragers
humans living creatures we can
really start to see that there has to be
some sort of a top-down organizing force
otherwise it would be like a land uh
slide or a lightning strike it would
just dissipate away
so he says look because the top-down
forces are so important
and over evolutionary time end up
shaping the parameters and the set of
what is possible or functional at the
lower level
it really needs to have an equal hand
causally so
we also need to look at this from a
top-down bottom-up perspective
lots of people have thought about these
types of issues in systems theories
and in other areas but the three areas
that um i just want to emphasize here
before bridging to active inference
are that there's a continuum of larger
and smaller things
that has to basically be true because
you can't have something bigger inside
of something smaller
and in the context of this debate that
we're framing
uh to get set up for which is between
internalism and externalism

you can think about things that are bigger as being more external and things that are smaller as being more internal so like the story about the smaller things is going into the organ into the cells into our cells but the story going out is about the ecosystem or the the constructed niche and these are things that are external and so that's gonna come back later in the paper and then also um as a consequence of that there's this nestedness or this embeddedness to these different levels of analysis and there's again many vocabularies for this type of recursive or embedded system and hopefully we'll address some but also feel free to provide feedback these are just some of the general aspects of multi-scale systems and some of the questions about multi-scale systems that the paper is going to address eventually and the field of research but also there's some philosophical work from areas other than the inactive cognitive area all right so the next topic is going to be about partitioning between internal and external cognition which is another way of saying what is the internalism versus externalism debate about how are we going to move beyond that how might the free energy principle be

involved in that
and in a sense the question that this is
asking
is how do we do causal modeling of
situations
where internal and external states
interact for example
through sense and action so let's think
about how sense and action are
interacting even at one level of
analysis what the ant
perceives for example in its actions
what you perceive and its actions
and then immediately one of the
questions that many people
have asked is okay so how do we make
that story
multi-level because the ant we could
simulate
it as just a robot with inputs and
outputs and sort of black box inside but
it
is also composed of the tissues and the
cells and it seems a little bit
ridiculous if we would have to simulate
from the cell
up to the ecosystem level so we have to
be able to
separate that model somewhere even if
only
for computational tractability or
comprehensibility so we got to make this
trade-off between
making causal models of systems that are
at one level of analysis
but including things that are at a
smaller level of analysis
that can have consequences that bubble

up for example a
drug binding to a receptor can bubble up
to behavior
that's like a story about upward
causation or someone's choice or
someone's context
could dictate their physiology through
any number of mechanisms which food or
molecules are available
so there is going to be times where that
cross-level interaction
is really important for the stories that
we tell for the narratives that how we
understand
the world um how we communicate about
those things so
how does the multi-level perspective
relate to causal modeling
if your story is about two people who
are fighting is the story only going to
be at that level or are you going to
bring in society are you going to bring
in the molecular
stuff how much upper and lower
context do you provide for causal
modeling at a given
level of analysis
how is causal modeling related to the
possible actions of a system
also known as the adjacent possible
for example what else could be and this
just kind of reminds me of how when
someone describes
a near-miss or a situation that could
have been another way
they're not saying it could have been
any other way
this thought experiment has been carried

out by gould
and others in the sense that yeah
something
happened a certain way which means it
was permitted by reality
and then you're envisioning a slightly
different scenario for example
the ball hits the other side of the roof
and it ends up rolling down a different
direction
doing some other outcome and the ball
still has the same physics and you're
still the same person so it's not any
alternate reality
it's a very tightly constrained causal
reality and what kinds of
alternate realities do people propose or
speculate
about ones that are within the causal
latent model of the world
where all stories where you cross the
freeway are dangerous for example
so that's the causal model that
underlies what we
genuinely generally are talking about
when we're talking about
what could happen we're not saying
anything could happen we're seeing a
pretty tight model
given about how we think about cause and
effect in the system
why is causal modeling so important to
active inference and there's two
roads into this i would say one
perspective
is why is causal modeling important from
the point of view the organism and
another

would be of the scientist or the
observer so
first from the point of the organism
there's a lot of ways to
frame exactly what it is an animal does
and a really
common area that people study this
without the
psycho baggage is in animal foraging
so whether it's a rodent or an ant
people or a bird people are really
interested
in the bioenergetics of and the
decision-making
that underlies how the animal uh goes
about searching for food
and how does that depend on the dynamics
of the ecosystem
and so one popular framework
in studying foraging is optimal foraging
theory
which you can just by the name of it
imagine is kind of like optimal
investing
strategy and there's a lot of variations
on it but it's just saying
what the ant is trying to do or what the
bird is trying to do
is given the costs and the benefits or
as estimated by the bird the predicted
costs and benefits
of the environment and the processing
time it's going to do this
reward maximization and
the good news about that approach is it
often works a lot of times animals are
near their predicted efficiency uh
however other times

it can be quite far off and it's not exactly apparent why that might be perhaps some constraint exists that was not modeled and so the reward maximization idea often succeeds or it captures a story or an important variable about a foraging system as you might expect but it doesn't really go the full way in explaining how the model is embodied or carried out by the animal because it doesn't necessarily have the kind of computational processing that a human simulation might not to compare the brain or the niche to a computer but just to say that that even as a physical computer imagined is still something that's quite different than the models that are being used to study the error divergence by which ultimately people are going to be asking whether the strategy was optimal in terms of its reward or not what active inference does is it says right the overall reward model is intractable it contains variables that are inaccessible or just not realistic for the nematode to be calculating for example in any way directly but what the animal can be doing is

engaging in this cycle between action
and perception and so it
is making these sensory hypotheses about
the world
embodied through action so for example
the eyes are circling around very
rapidly
where are they moving they're moving to
where your brain predicts that
you need to resolve your uncertainty
about your visual field
most so that's why even though your
actual visual resolution
on your retina or the density of the
cells the ability to perceive
the fine granularity of vision is quite
different across the back of your eye
yet the whole visual field unless you
have some sort of visual alteration
is uniformly clear and doesn't have a
blind spot so that's
evidence that at least visually we're in
a generative model
of what's happening and it shouldn't be
too far
away to say that that's what other
animals are also having at least
visually with a total secondary question
of whether it's experiential or not but
just visually
so active inference is a process theory
that
we can go into more another time that
has to do with
how organisms actually act in their
ecological niche
to resolve their uncertainty about the
kinds of problems they actually need to

resolve
therefore framing things like foraging
as a reward
obtaining but a precision maximization
behavior sequence and that framing turns
out to have
big implications so just wanted to frame
it that way
as far as the scientist's active
inference
the scientist also from an informational
perspective
has to think about which hypotheses
they're
experimenting or modeling with to reduce
their uncertainty
about what and just
as in active inference where the
generative model of the internal
state the organism in this example the
scientist
the internal model includes a generative
model of the latent causes in the world
and understanding the latent causes
better is what facilitates
uh better action and also better
experiments and
hopefully clearer transmission as well
how does the markov blanket idea come
into play
what pierces the blanket so here any
figure that you look at
of the markov blanket is going to have
uh basically the active inference
cycle with a markov blanket superimposed
down the middle
there's internal and external states and
then external states influence

sensory states sensory states influence

internal states

internal states have their own

autonomous or self-influencing dynamics

and through policy selection end up

implementing action states for example

the movement of an arm

action states influence the external

states the world states for example

which are also of course involved in

their own dynamics

the markov blanket is pierced on one

side by sense coming

in and by action on the other side going

out

and then the whole multi-scale question

is going to be about applying that

markov blanket all the way down idea

taking that seriously as far as

cognition one other area

that i thought could be kind of

interesting to relate this to would be

like other partitioning problems and

another partitioning problem that comes

up in

biological philosophy realms is this

question of nature versus nurture

and here i would point people towards

two

pretty recent and excellent books on the

left is beyond versus the struggle to

understand the interaction of nature and

nurture

by james tabery and then on the right

the mirage of a space between

nature and nurture by evelyn fox keller

and

they both have a different perspective

on the same question but
in in both cases there's this idea of
moving beyond
the binary partitioning of nature and
nurture and
so here in the ramstead 2019 paper
the framing of internalism and
externalism where they're sort of
entering into this stack is at the level
of cognition it's about cognitive
entities
whereas here's a partitioning problem in
the area of genetics
and in environmental and developmental
studies for example
and here's an example of where similar
internalism
internals the genes right smaller things
or externalism
outside bigger things the ecosystem is
seen as bigger than the person
with this implicit focused level of
analysis on the individual
and so um just to draw on evelyn fox
keller's book
she uses a metaphor of different kinds
of tasks and how
even if you just had two people so the
same type of agent
if you had two people building a brick
wall together you could at the end of
the day
say well this person added so and so
many bricks and this person added the
rest of the bricks
and so maybe you could say well 40 of
the work on this wall was performed by
that person

however imagine a situation where two people are needed to hold a hose to fill up a bucket for example even if one of the rules requires a specialty or more training or some other attribute even in that case that person can't do it alone so no matter how much water comes through that hose and it's all water coming through the hose you're never going to be able to say well 80 of the water was due to person a and 20 percent was due to person b so she uses that really clear metaphor to describe the relationship between nature and nurture how they're obviously related there is no such thing as a uh niche free organism especially when you think about this from an evolutionary ecology or ultimately active inference perspective organisms and their niche are what fitness is about that is what is being fit to what is the organism to their niche and you don't see things that aren't fit so it's unsurprising that we're finally attuned to things in our niche also there was a one interesting section in mirage of a space between where she traces the linguistic origins of the framing nature versus nurture because everyone has heard the phrase nature versus nurture is it nature or is it nurture

or it's in our dna it's in our nature
how did things come to be in this verses
sense that's also alluded to in the
title
of tabari's book and what fox
keller writes is recently a fashion has
arisen for tracing
the phrase nature and nurture and the
debate with which that phrase is
associated
back to shakespeare or at least to
prospero in the tempest
1623 who writes off caliban as
uneducable
quoting shakespeare a born devil on
whose nature
slash nurture will never stick
back to fox killer some have traced it
further
back to a monograph on children's
education written by an elizabethan
pedagogue
richard mulcaster mulcaster's words were
nature makes boy towered nurture sees
him forward
maybe it rhymes in another pronunciation
it's from 1581 and then
she concludes but in fact although both
molecaster
and shakespeare juxtapose the workings
of nature and nurture
neither invokes the term as an explicit
conjunction so not an or
they do not write of nature and nurture
as such
so they don't separate them and they
don't um make them
contra distincted versus each other

indeed when we look more closely at what
they do right we can see that their use
of the two terms does not invite
such a conjunction for there is no
presumption of an a priori disjunction
so what she's getting at there is
there really is no need to do an
argument against the concept of nature
and nurture
it's kind of like writing a hit article
on somebody who you disagree with
people who understand that there has to
be a move beyond
verses beyond an oppositional framing
of nature and nurture those people
should be and are making their positive
arguments
for why it should be framed a different
way
where the nature and the nurture are
things that are pulled out
or controlled for specifically and
transparently
in specific systems so that's an example
about how the philosophy of science
for example how we think about the
partition between internal and external
states
of phenotype relates to how we think
about multi-scale systems overall
because i think okay it reminds me so
much of this internalism and externalism
in cognitive sciences debate
reminds me a lot of this historical
debate in
genetics all right
next section perspectives on systems
modeling

and action and here the motivating question is about how do we carve nature at the joints which is a quote attributed to plato in the phaedrus allegedly and the quote alludes to this idea that nature whether one thinks that systems can be defined tightly and they're just on their own billiard ball trajectory or whether one has a relational perspective where there's just these multi-level networks of relations and being and becoming whatever it is at some point we want to carve it at the joints because otherwise we're going to get swept up in the storm and people will often hold that that's not their position or that they don't want to carve systems at the joints but i think if you're listening to this i hope you can see that at some point at least in action you have to be dealing with nature at the joints so to speak are they the right ones or not isn't that the whole question so one thing that is a connection is we can ask how is the fep the free energy principle or active inference going to help us frame this problem of carving nature at the joints this is kind of one of the big

ideas from a philosophical perspective
is about classification and about
distinguishing
things whether it's objects or types of
processes
or um ideas distinguishing
things and those are part of nature is
part of philosophy and so coming back to
this
essential question of carving nature of
the joint and then
asking well how can we use things like
fep
active inference to frame the problem
structurally
to give us in ontology and in fact in
the paper they'll talk about an ontology
and
subsequent work also um worked on
ontologies
so that's sort of the big motivation
it's why
this quote is being brought up in the
context of this paper we're about to
discuss
because the whole point is about us
carving nature at the joints
so let's go a little bit deeper into
systems modeling and carving nature at
the joints
and the question is how are descriptive
we'll just call that type 1
and action oriented type 2 models
related or different
and so just to explain what those 1 and
2 are
by descriptive model i mean a system
description only and so two

examples just to sort of pin down the
qualitative to quantitative
spectrum the first mention here is to
bruno latour's
actor network theory or ant and
it's an awesome piece of maybe
meta sociology not sure exactly what the
torah would call it
that just says the way to approach the
system is by
showing up finding out what are the
traces and
being open with respect to which actors
are involved in which processes through
which
networks of influence so it's a very
unbiased
description or moving towards unbiased
not to step into that sort of uh
trap i guess and then um
it's very qualitative it's relational
it's also about
the observers knowing and relationship
with the system
not just the way that aspects of the
system are associated with each other so
often those things are left out in
quantitative or in some scientific
world use and on the other end of the
system's description
only spectrum i've pointed to
a package of complex systems
design tools called cadcad now cad cad
goes beyond
systems description it turns out to also
be able to do things like
simulation and and other
more dynamical approaches like signal

processing

but what i mean by putting cad cad here

is that it can be used

purely depictively to specify a complex system

especially one that's computational so there might be

a question when you're actually trying

to model a organic ecosystem

of how do you measure some every single

uh you know piece of the ecosystem and

how many variables is enough to describe

the fineness of the temperature gradient

or something like that

but when the simulation is about a

digital system

it can be more deterministically and

controllably

simulated if one can imagine so there's

one type of world modeling related to

system description

but that obviously enables simulation

and inference as well

the second kind of models are

action-oriented models

so action-oriented models are those

where

ultimately we're just we're assigning

meaning to states

voting with your fee voting with your

dollars it's

when we want to have effective action or

at least

the system acts like it wants to

something like maybe what dennett would

call the intentional stance

so the descriptive modeling might be a

question that you're asking the computer

like given this scatter of points which axes capture the most statistical variants and that would be just it running some sort of algorithm to find the maximum variance explained whereas a another type of algorithm like more of a type 2 action-oriented algorithm might be learn the relationship between this input and the optimal performing behavior in the output so that's what action-oriented models are about next question how does the multi-level perspective influence our answers above so if we're going to have a story where there's nature's being carved at its joints maybe around the markov blankets maybe with scents coming in and action coming out even if we have that at one level so we have an asian-based model of the ants that are bumping into each other and at one level the sense of one is the action of another and so we have a sort of bumper car model um where does the multi-level perspective come into play and that's kind of thinking about the bumper car one it's where you would model first the cars bumping into each other and then you could go into the engine or you could go out to the power grid and to the people who are riding on the bumper cars or something like that so

how are we going to deal with the
complexity of causal stories
relationships
some of what might which uh be really
important
and like the example of a molecule that
binds to a receptor
others which might be spurious and the
reality is that because of the
combinatorics
most of the options across levels will
be spurious
so most of the ecological relationships
can't be for example
one-to-one because there's so many that
are just bumping into each other all the
time
just to give one example from ecology so
how are we going to carve nature at its
joints
do it cleanly are we trying to represent
stuff like one are we trying to act like
two
do one and two fall out as a function of
doing one do we get
one and two as a function of following
out just two
how does emergence and emergent
properties fit into all of this
so by emergence or emergent properties
people often mean
a sort of system outcome where there's
some property or characteristic
or attribute that is arising at a higher
level
again bigger broader spatial and
temporal scales
based upon interactions or some sort of

interfacing

of the smaller level so sometimes these

emergent properties

using the classification here of badao

are

in the weak category and in the strong

category it's not the only way to break

down

immersions not to only take here but i

find this one pretty helpful

first strong emergence badao equates

with basically magic

you can read the papers more if you want

to

explore more nuance but the strong

emergence is from

something that's totally disjoint to

something that's

has a total integrity like language or

awareness for example and then weak

emergence is contrasted and that's most

of the

physical systems where even if the

relationships between subunits are

really nuanced

or the computational uh capacity

required to simulate the system would be

extreme

those systems are still weakly emerging

in that it still just is ants bumping

into each other

now they're developing and that's sort

of crazy and evolution shapes that and

everything

but still we're talking about a physical

system not in the

genre for example of um awareness

arising spontaneously so how is that

going to play into
uh the discussion with active inference
showing up on the scene
where will this relate to immersions
these are things
anyone's welcome to chime in on
also we can tie the idea of emergence
and the multi-scale systems approach
to this debate that we often hear about
reductionism versus holism so
reductionism
is simply a method of explanation
where bigger things are made sense of by
appeals to the properties of their
smaller things
like the bigger wall is um orange
because each of the smaller
pixels is orange or something like that
there's a variety of reductionist
explanations
and it's just one of our hands it's one
of our approaches or our wings
holism is kind of flipping that
uh schema of reductionism saying
actually smaller things
are made sensitive in terms of the
bigger thing
and that um is kind of the opposite
edge viewpoint and so we have some
sliding to
toward a holistic understanding some
sliding towards a reductionist
understanding a lot of
philosophical work in this area and the
thing that we would just want to note
here would just be
first ask continually how is the free
energy principle or active inference

relating to
reductionism versus holism because
sometimes
it may feel like there is a sort of a
philosophy of
science shift associated with taking on
the active inference framework
and so what are the implications of
taking that seriously where does that
put this reductionism versus holism
is it going to be put into a similar bin
as nature versus nurture
as kind of a turn of phrase or a false
dichotomy
that was just eventually seen through or
will there be people who still think
that that sort of binary phrasing
whichever side they find themselves on
is going to be appropriate and again
this is where dennis noble's
work on the bottom up and top down is
related
the other work here is girdle escherbach
which is the 1979 book
by douglas hofstetter and here's one of
the figures from the book where it says
uh mu mu i don't know how you'd say it
which is a allegorically at least
a sort of innovative response or a
canonical response
depending on who you ask of zen koan
and zen is a topic that's brought up in
the book
and the m is written
of letters that are containing smaller
letters that say reductionism
and then the u is made up of letters
that

spell out the other words so it's kind
of this multi-scale um
thought loop really captures the spirit
of the book well
and also it captures at least informally
through art as often is the case what it
looks like
to go beyond the reductionism versus
holism debate so first if only in a text
illustration but then eventually
how does that relate to a computer
screen how about a person interacting
with a computer screen
how about two people interacting with
their computer screens
all right few more perspectives on
systems modeling and action
how is world modeling related to
pragmatics and action
so the questions to ask here is really
is there a best map
of the territory uh short answer no
and this is true for geospatial maps as
well as
fields of affordances which we talked
about in the previous active streams
which also often demand to be mapped
for example skill set mappings or
concept mappings
so there's a lot to say about map and
territory of course
but for here it's suffices to say that
the best map
is defined by the mission by the context
by the degree of certainty by all these
other features and factors related to
the user
and the situation so we're talking about

making maps

uh of meaning maps for the world we're
talking about doing world modeling for
effective inference and action as we
were talking about previously

so here's just one quote from the paper
to

motivate you towards the things that you
might be able to hear them

expand more on they write the

multi-scale free energy principle model

allows us to cast the boundaries of

cognition as assembled and maintained in
an informational dynamics

across multiple spatial and temporal
scales

crucially we shall show that this

multi-scale application

of the fep free energy principle implies

both ontological

and methodological pluralism so a few

thought questions whether this is before

or after reading the paper

you could ask is the free energy

principle

pluralistic with respect to non free

energy principle theories

or frameworks or principles which

levels or types of theory is the fep

consistent or discordant with and one

starting point there would be

searching for carl fristen's interview

in the 2018

alias bulletin and it's an interesting

question to ask about any theory is okay

well who is this

changing the mind of or what is the

opposite of this theory and

is it true or does anyone hold the true
opposite
of this theory does anyone hold a 50
opposite is it 50 by outcome
or is it fifty percent by axiom in that
they diverge
so you can always ask that and i found
that that was a
very um interesting route to go down as
far as the fep
another quote from the paper is just
because it's such a
direct claim our claim is that
which among these which are the situated
barriers of
talking of systems which among these are
the most relevant will depend on the
phenomena being studied and the
explanatory interests of researchers
and so there's a few notions like
markov blanket which we talked about
earlier is that sense in action
piercing that's related to ecological
psychological concepts as well
the markov blankets
are defined in a few different ways
they're defined
by statistical covariations but
one type of those is going to be causal
and also another type of those could be
like a measurement
and so the kinds of measurements that we
make the types of causal relationships
that are important and then which models
we apply to them
is going to depend on uh what is enacted
by the researchers because they're also
part of embodied and cultured cognition

so briefly what are mythological and
ontological pluralism so
methodological pluralism means using a
variety of methods or ways of doing
ontological pluralism means having a
variety
of ontologies or ways of knowing
and pluralism is referring to there
being multiple
now there are other schools of thought
that are sort of higher order than
pluralism as to
how do you cut through that is it just
theory anarchy like um
fair bend suggested where the pluralism
can just be taken to this extreme
uh equivalent to almost a novelty search
in theory space
or is there going to be some way to cut
through that and determine
which of the pluralistic options or
possibilities or actualities
is preferable because ultimately we do
have to act that's that type 2 type of
model
so ask yourself for direct action
or for more philosophical topics what is
utilitarianism or pragmatism
how do such ideas relate to pluralism so
it's one thing to say we're scientists
and whichever level of analysis
is most relevant well that's going to
depend on what we're studying
and what we're interested in explaining
but
should it be that really the best way to
approach these systems
should we really um knowing that we can

never

remove for example individual bias

but still wanting to model systems so
that we can act effectively

how are we going to balance just this
sort of ruthless utilitarianism

pragmatism whatever ends justify the
means

um whatever preconceived worldview

whatever your bayesian prior is just
stick with that no matter what happens

how how is that related to cutting
through

the noise of pluralism

these are again just questions to raise
in active inference

how are ideas related to action and what
kind of ideas or mental states are there

there's a few kind of interesting
categories that we perceive at least
experientially various schools of
thought

on that but where are we going to fit
these into the active inference

framework are those just different loops
in our generative model things like
emotion

or desires or dreams or secrets

where are those going to fit into this
active inference

framework how integrated is that going
to be across

types of ideas

and how are our conserved or novel ideas
respectively meaning orthodoxy or
heterodoxy

associated with the expression of
conserved or novel actions

so that is ortho practice or
heteropraxis
and linking that to active inference
which is kind of
just framing this two-stroke engine or
this continual ooda
loop a lot of ways to think about it
it's just framing this continual flow
between
action and inference so how are those
two linked here's another way of asking
it
and just to conclude with two
books to study how this has been studied
from a philosophy of science perspective
on the right is scientific pluralism uh
edited by keller longino
and waters and then on the left is a
book called studying human behavior
how scientists investigate aggression
and sexuality by helen longneau
and these books talk about the
question of pluralism in science and
different ways that people have
sort of cut through the massive theories
or not
or contributed to it i guess in some
cases and there's just a really
interesting
body of thought and so it's uh useful to
think
for people who are in the free energy
space
as well as from the outside what are the
bridges that link
what's happening with the free energy
principle and active inference
to more classical philosophy of science

questions

and so hopefully in the first in

alias paper from 2018 and

also some other excellent work from the

last few years

we'll come to a better understanding of

that so

thanks for participating which i guess

was me and you as the listener

we will provide a follow-up form to live

participants

and we would welcome any feedback

suggestions

questions you have and yeah

stay in communication with us so

thanks again for listening and

we hope to either see you at 5.1 or

a future discussion